The conditions of the definition can be formulated in 4 steps :

(i) Evaluate f at x_0 and $x_0 + \Delta x$: $f(x_0)$ and $f(x_0 + \Delta x)$

(ii) Compute Δy : $\Delta y = f(x_0 + \Delta x) - f(x_0)$

(iii) Divide Δy by Δx $(\neq 0)$: $\dfrac{\Delta y}{\Delta x} = \dfrac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$

(iv) Compute the limit as $\Delta x \to 0 (\neq 0)$: $m_{\tan} = \lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x}$.

The computation of the slope of the graph in the Illustration 10.1 can be facilitated using the definitions.

(i) $f(1) = 1^2 = 1$. For any $\Delta x \neq 0$

$$f(1 + \Delta x) = (1 + \Delta x)^2 = 1 + 2\Delta x + (\Delta x)^2$$

(ii) $$\Delta y = f(1 + \Delta x) - f(1) = 2\Delta x + (\Delta x)^2 = \Delta x\,(2 + \Delta x)$$

(iii) $$\frac{\Delta y}{\Delta x} = \frac{\Delta x(2 + \Delta x)}{\Delta x}$$
$$= 2 + \Delta x\,.$$

(iv) $$m_{\tan} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$
$$= \lim_{\Delta x \to 0} (2 + \Delta x) = 2 + 0 = 2\,.$$

Example 10.1

Find the slope of the tangent line to the graph of $f(x) = 7x + 5$ at any point $(x_0, f(x_0))$.

Solution

Step (i) $f(x_0) = 7x_0 + 5$.

For any $\Delta x \neq 0$,

$$f(x_0 + \Delta x) = 7(x_0 + \Delta x) + 5$$
$$= 7x_0 + 7\Delta x + 5$$

Step (ii)
$$\Delta y = f(x_0 + \Delta x) - f(x_0)$$
$$= (7x_0 + 7\Delta x + 5) - (7x_0 + 5)$$
$$= 7\Delta x$$

Step (iii)
$$\frac{\Delta y}{\Delta x} = 7$$

Thus, at any point on the graph of $f(x) = 7x + 5$, we have

Step (iv)
$$m_{\tan} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$
$$= \lim_{\Delta x \to 0} (7)$$
$$= 7.$$

Note that for a linear graph, $\dfrac{\Delta y}{\Delta x}$ is a constant, depends neither on x_0 nor on the increment Δx.

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